

# **BSBINS401 - Analyse and Present Research Information**

## **Session 1: Introduction to the Course and Data Science**

Lecturer: Jordan Hill

## Learning Objectives

- **Understand the course structure and expectations.**
- **Explore the role of data science in various industries.**
- **Familiarize with unit requirements and assessments.**
- **Initiate application for Google Cloud Credits.**

# Welcome to the Course!

- **Course Code:** BSBINS401
- **Course Title:** Analyse and Present Research Information
- **Focus:** Introduction to Data Visualization Tools using Python



## Course Overview

- **Duration:** 18 weeks
- **Delivery Mode:** Face-to-face learning
- **Primary Tools:**
  - Python
  - Jupyter Notebooks
  - Kaggle
  - Matplotlib and Seaborn libraries

## Course Structure

- **First Cycle:**
  - Work with a standard dataset provided.
  - Conduct research, analyze data, and present findings.
- **Second Cycle:**
  - Scope your own research project.
  - Find or collect your own dataset.
  - Repeat the cycle independently.

## Assessments Overview

| Assessment                                                         | Description                                                                                  | Due Date |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------|
| <b>Assessment 1:</b> Research and Presentation on Standard Dataset | Analyze provided dataset and present findings through a report and presentation              | Week 7   |
| <b>Assessment 2:</b> Independent Research Project                  | Conduct independent research and present findings in a comprehensive report and presentation | Week 16  |
| <b>Assessment 3:</b> In-Class Closed Book Exam                     | Complete closed book exam to assess understanding of research principles and methodologies   | Week 17  |

## Learning Outcomes

By the end of this course, you will be able to:

- **Identify research requirements and objectives.**
- **Collect, organize, and present research information.**
- **Maintain information securely according to policies.**
- **Prepare and produce research reports.**

# Role of Data Science

- **What is Data Science?**
  - Interdisciplinary field combining domain knowledge, programming skills, and math/statistics.
- **Data Science in Industries:**
  - **Healthcare:** Predictive analytics for patient care.
  - **Finance:** Fraud detection and risk management.
  - **Retail:** Customer segmentation and recommendation systems.
  - **Transportation:** Route optimization and autonomous vehicles.



## Why Data Visualization?

- **Understand Complex Data**: Visual representations make patterns and trends easier to comprehend.
- **Communicate Insights**: Effectively share findings with stakeholders.
- **Facilitate Decision Making**: Support data-driven decisions in organizations.

## Tools We'll Use

- **Python**: Versatile programming language.
- **Jupyter Notebooks**: Interactive computing environment.
- **Kaggle**: Platform for data science competitions and datasets.
- **Matplotlib and Seaborn**: Libraries for creating static, animated, and interactive visualizations.

## What is Kaggle?

- **Kaggle.com**: A platform offering datasets, competitions, and a community of data scientists.
- **Features**:
  - Access to public datasets.
  - Online coding environment.
  - Learning resources and tutorials.

## Cloud Credits for Education

**Why Google?:** Google Cloud is the backbone of Kaggle— so if we can get some credits it will give us some extra flexibility with running our workloads.

**Why AWS?:** AWS has a number of great integrations with the open source AI & Machine learning community. For example, Huggingface, via AWS SageMaker.

**Why Azure?:** Azure has a number of great integrations with the open source AI & Machine learning community. For example, Huggingface, via Azure ML.

These three are the major players when it comes to AI & cloud computing. If you continue in this field you will likely need some form of support from these providers.

# Applying for Google Cloud Credits

- **Benefits:**
  - Access to cloud computing resources.
  - Ability to run computing-intensive tasks.
- **How to Apply:**
  - Visit [Google Cloud for Education](#).
  - Follow instructions to redeem credits.

# AWS Educate

- **AWS Educate :**
  - Access to cloud computing resources.
  - Ability to run computing-intensive tasks.
- **How to Apply :**
  - Visit [AWS Educate](#).
  - Follow instructions to redeem credits.

## Activities for Today

- **Introductions :**
  - Share your background and interests.
  - Discuss what you hope to learn from this course.
- **Case Studies :**
  - Explore examples of data science impacting industries.
  - Discuss the societal implications of data science.
- **Set Up Accounts :**
  - Create accounts on Kaggle and Google Cloud.
  - Apply for Google Cloud Credits.

# Getting Started with Python

- **Check Installation :**
  - Ensure Python and Jupyter Notebook are installed on your computer.
  - Alternatively, set up a cloud environment via Kaggle or Google Colab.
- **First Notebook :**
  - Create a simple "Hello, World!" notebook.
  - Explore basic Python commands.



# Unit Requirements

- **Performance Evidence :**
  - Research, analyze, and present findings on at least two occasions.
  - Align with organizational requirements.
- **Knowledge Evidence :**
  - Understand organizational policies.
  - Recognize reliable and valid research practices.
  - Familiarity with research strategies and information sources.

# Policies and Procedures

- **Confidentiality and Privacy:**
  - Adhere to privacy laws and organizational policies.
  - Secure handling of data.
- **Data Security:**
  - Best practices for storing and accessing information.
- **Academic Integrity:**
  - Avoid plagiarism.
  - Proper citation of sources.

## Support and Resources

| Lecturers   |                                                                                | Office Hours         | Resources                                                                                                                  |
|-------------|--------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------|
| Jordan Hill | <a href="mailto:jordan.hill@nmtafe.wa.edu.au">jordan.hill@nmtafe.wa.edu.au</a> | Mon-Fri, 9 AM - 5 PM | <ul style="list-style-type: none"><li>• Course materials on LMS</li><li>• Access to library and online databases</li></ul> |
| Hong Fu     | <a href="mailto:hong.fu@nmtafe.wa.edu.au">hong.fu@nmtafe.wa.edu.au</a>         |                      |                                                                                                                            |

## Out-of-Class Activities

- **Required Reading:**
  - ["What is Data Science?"](#)
  - *Python for Data Analysis* by Wes McKinney, [Chapter 1: Preliminaries](#)
- **Tasks:**
  - Ensure Python and Jupyter Notebook are set up.
  - Familiarize yourself with basic Jupyter Notebook functionality.



## Any Questions?

- **Open Floor:**
  - Any questions about the course structure?
  - Concerns or topics you'd like to explore?
- **Contact Us:**
  - Feel free to reach out via email or during office hours.

## Next Session Preview

- **Topic:** Data Visualization Basics and Tools Overview
- **What to Expect:**
  - Learn about different types of graphs.
  - Introduction to Matplotlib and Seaborn.
  - Hands-on activities with Jupyter Notebooks.
- **Preparation:**
  - Complete the required readings.
  - Bring your laptops for in-class exercises.